

EduGenie – Data Flow Diagram

Project: EduGenie – Google Gemini Powered Learning Assistant

1. Introduction

The Data Flow Diagram (DFD) shows how information moves through the EduGenie system from the student to the AI service and back to the student.

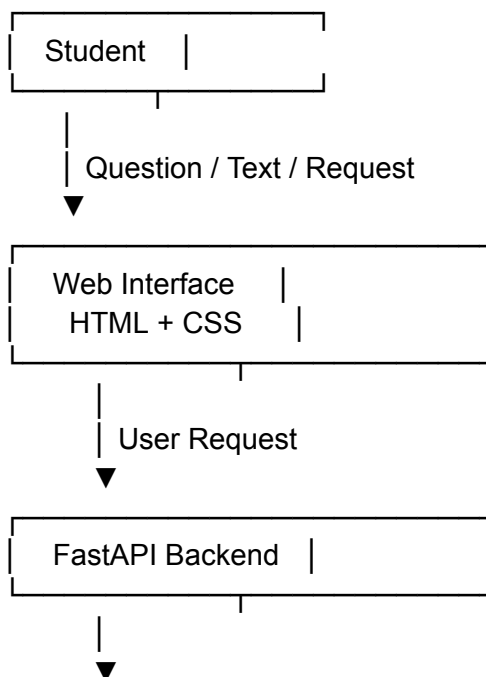
2. Level 0 Data Flow

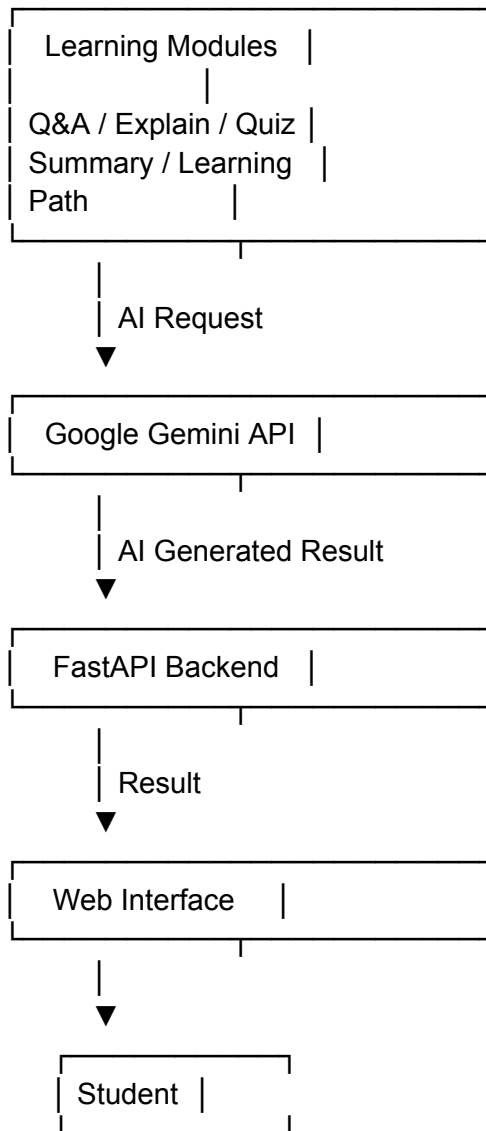
The main data flow is:

Student → EduGenie System → Google Gemini API → EduGenie System → Student

The student provides a question, educational text, or learning request. The EduGenie system processes the request and uses the Google Gemini API when AI-generated processing is required. The generated result is then displayed to the student.

3. Level 1 Data Flow





4. Data Flow Description

1. The student enters a question, text, or learning request.
2. The web interface sends the request to the FastAPI backend.
3. The backend identifies the required learning function.
4. The appropriate learning module processes the request.
5. The module communicates with the Google Gemini API when required.
6. The generated response is returned to the backend.
7. The result is displayed to the student through the web interface.

5. Conclusion

The Data Flow Diagram represents the movement of user input and AI-generated output through the major components of EduGenie.